

SOFTWARE REQUIREMENTS SPECIFICATION

Project Name: Digi Keralam 2.0

Version: 1.0 Prepared By: KSITM Business Analyst Team **Date:** February 4, 2026

Table of Contents

1. **Introduction** 1.1 Purpose 1.2 Document Conventions 1.3 Intended Audience and Reading Suggestions 1.4 Product Scope 1.5 References
 2. **Overall Description** 2.1 Product Perspective 2.2 Product Functions 2.3 User Classes and Characteristics 2.4 Operating Environment 2.5 Design and Implementation Constraints
 3. **System Features (Functional Requirements)** 3.1 Landing Page & Content Delivery 3.2 Registration & Profile Management 3.3 Administrative Management (KSITM & LSGD) 3.4 Trainer Hierarchy & Resource Allocation 3.5 Session Management & Location Logic
 4. **External Interface Requirements** 4.1 User Interfaces 4.2 Hardware Interfaces 4.3 Software Interfaces
 5. **Non-Functional Requirements** 5.1 Performance 5.2 Safety & Security 5.3 Reliability
 6. **Appendix A: Glossary**
-

1. Introduction

1.1 Purpose

The purpose of this SRS is to describe the software requirements for the **Digi Keralam 2.0** platform. This web-based application aims to bridge the digital divide by educating the general populace of Kerala on current technologies. The document details the system's behavior, user roles, administrative hierarchies, and location-based session management.

1.2 Document Conventions

- **Bold text** is used for emphasis and for highlighting specific roles (e.g., **Super Trainer**) or system entities.
- **"Shall"** indicates a mandatory requirement.

- **"Should"** indicates a desirable but not mandatory requirement.

1.3 Intended Audience

This document is intended for:

- **Project Managers (KSITM):** To ensure the project aligns with the state's digital literacy goals.
- **Development Team:** To understand the functional logic, specifically the location-based mapping.
- **QA Team:** To design test cases for hierarchy validation and session creation.

1.4 Product Scope

Digi Keralam 2.0 is an educational management system. It functions similarly to a Learning Management System (LMS) but with a strict geographical and hierarchical backbone. The system focuses on three pillars:

1. **Safe Tech Campaign:** Cyber safety and ethics.
 2. **AI Education:** Fundamental AI literacy.
 3. **DEED (Digitalization of Essential Documents):** Training on digital documentation.
-

2. Overall Description

2.1 Product Perspective

This system is a standalone web application that will interface with existing government databases for location verification (Local Self Government Department data) where applicable. It serves as the primary interface for both the administration (KSITM/LSGD) to manage resources and for the public to access training.

2.2 Product Functions

- **Public Awareness:** Hosting introductory content via a mascot-led video interface.
- **User Segmentation:** Categorizing citizens by proficiency (Beginner, Intermediate, Expert).
- **Hierarchical Management:** Managing a multi-tier trainer structure (Super Trainer → Master Trainer → Trainer).
- **Geo-Fencing:** Strictly mapping sessions and trainers to specific Panchayats, Municipalities, Corporations, and Wards.

2.3 User Classes and Characteristics

1. **Citizen (End User):** Varies from students to senior citizens. High variance in technical literacy. The UI must be extremely intuitive.
2. **Trainer:** Field staff. Requires mobile-friendly access to view assigned sessions.
3. **Master Trainer:** Mid-level manager. Needs dashboard capabilities to view ward-level data and assign trainers.
4. **Super Trainer:** High-level manager. Oversees multiple Master Trainers.
5. **LSGD Admin:** Local Government staff. Responsible for verifying trainer identities and assigning roles.
6. **KSITM Admin:** Super Administrator.

2.4 Operating Environment

- **Client Side:** Modern web browsers (Chrome, Firefox, Safari, Edge) on Desktop and Mobile devices.
- **Server Side:** Cloud-hosted environment scalable to handle concurrent state-wide access.

3. System Features

3.1 Landing Page & Content Delivery

3.1.1 Description The entry point for all users, designed to be welcoming and educational.

3.1.2 Functional Requirements

- **REQ-LP-01:** The homepage shall feature a high-quality video player prominently displaying the "Mascot Introduction" video.
- **REQ-LP-02:** The navigation bar shall be persistent and contain links to: Home, Safe Tech, AI Module, DEED Module, Sign In, and Register.
- **REQ-LP-03:** The "About" section shall describe the three pillars (Safe Tech, AI, DEED) in simple language.

3.2 Registration & Profile Management

3.2.1 Description The system must handle two distinct registration workflows with different data requirements.

3.2.2 Functional Requirements

- **REQ-REG-01 (Citizen):** The system shall collect: Name, Age, Mobile Number, Email (optional), and Residential Location (District/Local Body).
- **REQ-REG-02 (Proficiency):** During registration, Citizens must self-select a proficiency level: *Beginner, Intermediate, or Expert*.
- **REQ-REG-03 (Trainer):** Trainer registration must collect detailed location data: District, Local Body Type (Panchayat/Municipality/Corporation), Local Body Name, and Ward Number.
- **REQ-REG-04:** Newly registered trainers shall be placed in a "Unverified/Pending" pool visible only to LSGD Admins.

3.3 Administrative Management (KSITM & LSGD)

3.3.1 Description The backend management structure for the platform.

3.3.2 Functional Requirements

- **REQ-ADM-01 (KSITM):** The KSITM Admin shall have the ability to create and manage LSGD Admin accounts for each district or region.

- **REQ-ADM-02 (LSGD):** The LSGD Admin shall have a "Trainer Management" dashboard.
- **REQ-ADM-03 (Role Assignment):** LSGD Admins shall select users from the "Pending Trainer" pool and assign them a specific role:
 - **Super Trainer:** Assigned to a cluster of regions.
 - **Master Trainer:** Assigned to specific local bodies.
 - **Trainer:** Assigned to specific wards.

3.4 Trainer Hierarchy & Resource Allocation

3.4.1 Description Defining the command chain for field operations.

3.4.2 Functional Requirements

- **REQ-HIER-01 (Super Trainer):** A Super Trainer shall be linked to Master Trainers (configurable).
- **REQ-HIER-02:** The Super Trainer dashboard shall display aggregated data: Total Wards Covered, Total Sessions Conducted, and a list of subordinate Master Trainers.
- **REQ-HIER-03 (Master Trainer):** The Master Trainer shall have the authority to view the schedule of all Trainers within their jurisdiction.

3.5 Session Management & Location Logic

3.5.1 Description The core "product" delivery mechanism—scheduling and managing classes.

3.5.2 Functional Requirements

- **REQ-SESS-01:** Master Trainers shall have a "Create Session" interface.
- **REQ-SESS-02:** When creating a session, the Master Trainer must define:
 - **Category:** Safe Tech, AI, or DEED.
 - **Target Audience:** Beginner, Intermediate, Expert.
 - **Location:** Specific Panchayat/Municipality/Corporation and Ward.
 - **Mode:** Online or Offline (Physical Venue).

- **REQ-SESS-03 (Allocation):** The system shall filter available Trainers based on the session's location (Ward/Panchayat) and allow the Master Trainer to assign a specific Trainer to the session.
 - **REQ-SESS-04 (Citizen View):** Citizens shall only see sessions relevant to their registered proficiency level and their selected location.
-

4. External Interface Requirements

4.1 User Interfaces

- **Front-end Standard:** The UI shall be responsive, adhering to WCAG 2.1 accessibility standards to support senior citizens and users with disabilities.
- **Language:** The interface shall support both Malayalam and English.

4.2 Software Interfaces

- **Database:** PostgreSQL or MySQL for relational data management of users and location hierarchies.
 - **Authentication:** OAuth 2.0 or OTP-based authentication for mobile number verification.
-

5. Non-Functional Requirements

5.1 Performance

- **Response Time:** 95% of web pages shall load within 2 seconds over a standard 4G connection.

- **Capacity:** The system shall support up to 10,000 concurrent users during peak registration drives.

5.2 Security

- **Data Privacy:** Citizen data shall be encrypted at rest.
- **Access Control:** Vertical Privilege Escalation protection must be in place (e.g., a Trainer cannot access the LSGD dashboard via URL manipulation).

5.3 Reliability

- **Availability:** The system shall have a 99.9% uptime during business hours (8 AM - 8 PM).

Appendix A: Glossary

- **KSITM:** Kerala State IT Mission.
- **LSGD:** Local Self Government Department.
- **Ward:** The smallest administrative unit in the local governance structure.
- **LLM (in context):** Refers to the project's similarity to a "Large Language Model" in terms of being a repository of knowledge, not necessarily the technology stack itself.